

MTH 203: Sequences and series

Mid-examination 2: 11/10/2025,

Time: 12:00 to 13:00

Total points: 25

Problem 1. (1 point each) State whether the following statements are **true** or **false**. (No proofs required.)

- (a) Every subsequence of an unbounded sequence is divergent.
- (b) Suppose $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ are sequences such that $\lim_{n \rightarrow \infty} a_n = 1$ and $(b_n)_{n \in \mathbb{N}}$ is divergent. Let $(c_n)_{n \in \mathbb{N}}$ be the sequence defined by $c_n = a_n b_n$ for all n . Then $(c_n)_{n \in \mathbb{N}}$ is a divergent sequence.
- (c) The series $\sum_{n=1}^{\infty} x^n$ converges if and only if the sequence $(x^n)_{n \in \mathbb{N}}$ is convergent.
- (d) If $(a_n)_{n \in \mathbb{N}}$ is a bounded sequence such that $a_n \geq 0$ for all n , then it is necessarily a convergent sequence.

Solution.

Part (a): Answer: False

Consider the sequence $(x_n)_{n \in \mathbb{N}}$ defined by $x_n = (-1)^n$. This is a divergent sequence (see Lecture 8, Example 1). However, the subsequence $(x_{2n})_{n \in \mathbb{N}}$ is just the constant sequence

$$1, 1, 1, \dots$$

which is convergent.

Part (b): Answer: True

We have $b_n = c_n/a_n$ for any n such that $a_n \neq 0$. If we had $a_n \neq 0$ for all n , we would have this equality for all n . Since the $\lim_{n \rightarrow \infty} a_n \neq 0$, we can then just conclude that the sequence $(c_n/a_n)_{n \in \mathbb{N}}$ is convergent. (See Lecture 10, Corollary 2.)

However, the conclusion remains valid even if we do not have $a_n \neq 0$ for all n . Indeed, we can only have $a_n = 0$ for *finitely many* values of n since $\lim_{n \rightarrow \infty} a_n \neq 0$. Dropping a few initial terms does not affect the limit of a sequence. (This could be written more rigorously, but we will not do so here.)

Part (c): Answer: False If we take $x = 1$, the sequence $(x^n)_{n \in \mathbb{N}}$ is convergent since it is the constant sequence. But the series $\sum_{n=1}^{\infty} x^n$ is divergent.

Part (d): Answer: False Consider the sequence $(a_n)_{n \in \mathbb{N}}$ defined by

$$a_n = \begin{cases} 0 & \text{if } n \text{ is even,} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

Then $(a_n)_{n \in \mathbb{N}}$ is a bounded, divergent sequence such that $a_n \geq 0$ for all n . $\square$

Problem 2. (3 points each) Calculate the following:

(a)

$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n} - 1}{\sqrt{n} + 1} \right)$$

(b)

$$\sum_{n=1}^{\infty} \frac{2^{3n}}{3^{2n}}$$

You must justify all your calculations.

Solution.

Part (a): We have

$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n} - 1}{\sqrt{n} + 1} \right) = \lim_{n \rightarrow \infty} \frac{\sqrt{n} + 1 - 2}{\sqrt{n} + 1} = \lim_{n \rightarrow \infty} \left(1 + \frac{2}{\sqrt{n} + 1} \right).$$

We show that

$$\lim_{n \rightarrow \infty} \frac{2}{\sqrt{n} + 1} = 0.$$

Let $\epsilon > 0$ be given.

We choose N to be such that $N > (2/\epsilon - 1)^2$ (using the Archimedean property). Then, if $n > N$, we have $n > (2/\epsilon - 1)^2$. Thus, $\sqrt{n} > 2/\epsilon - 1$ and so $\sqrt{n} + 1 > 2/\epsilon$. Since $\epsilon > 0$ and $\sqrt{n} + 1 > 0$, this gives $\epsilon > 2/(\sqrt{n} + 1)$.

Thus,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{2}{\sqrt{n} + 1} \right) = \lim_{n \rightarrow \infty} 1 + \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n} + 1} = 1 + 0 = 1.$$

Remark: Simply writing something like

$$\lim_{n \rightarrow \infty} \frac{2}{\sqrt{n} + 1} = \frac{\lim_{n \rightarrow \infty} 2}{\lim_{n \rightarrow \infty} \sqrt{n} + 1} = 0$$

along with some justification like $\lim_{n \rightarrow \infty} (\sqrt{n} + 1) = \infty$ is not an acceptable answer at this point in the course since we have not discussed what it means for a sequence to “diverge to infinity”, etc.

Part (b): Since $2^{3n}/3^{2n} = (8/9)^n$, the series is just $\sum_{n=1}^{\infty} (8/9)^n$. Since $8/9 < 1$, this series converges. $\square$

Problem 3. (5 points) Let $(a_n)_{n \in \mathbb{N}}$ be a sequence such that $\lim_{n \rightarrow \infty} a_n = 0$. Prove that there exists a subsequence $(a_{n_k})_{k \in \mathbb{N}}$ of this sequence such that the series $\sum_{k=1}^{\infty} a_{n_k}$ converges.

Solution. We first choose a strictly increasing sequence of positive integers $(m_k)_{k \in \mathbb{N}}$ such that $|a_{m_k}| < 1/2^k$ for all $k \in \mathbb{N}$. This sequence is constructed by an inductive process.

First, we choose m_1 to be an integer such that $|a_{m_1} - 0| < 1/2$. (This is possible since $\lim_{n \rightarrow \infty} a_n = 0$.)

Now, suppose that for some integer $k \geq 1$, we have chosen $0 < m_1 < \dots < m_k$ such that $m_i \leq 1/2^i$ for $1 \leq i \leq k$. There exists an integer N such that for any $n > N$, we have $|a_n - 0| < 1/2^{k+1}$. Choose $m_{k+1} = \min\{m_k, N\} + 1$. We obviously have $m_{k+1} > m_k$. Since $m_{k+1} > N$, we also have $|a_{m_{k+1}}| < 1/2^{k+1}$. This completes the induction step.

This process gives us the sequence of integers $(m_k)_{k \in \mathbb{N}}$ as claimed.

Now, let us consider the subsequence $(a_{m_k})_{k \in \mathbb{N}}$. We have $|a_{m_k}| < 1/2^k$ for all k . We would like to use the Comparison Test (see Theorem 6 in Lecture 12) to conclude that the series $\sum_{k=1}^{\infty} a_{m_k}$ converges. Unfortunately, all the terms a_{m_k} may not be non-negative. To rectify this issue, we need to refine it further.

Let $A = \{m_k | a_{m_k} \geq 0\}$ and $B = \{m_k | a_{m_k} < 0\}$. Then, as $A \cup B$ is an infinite set, at least one of the two sets A or B must be infinite.

Suppose A is an infinite set. Suppose $n_1 < n_2 < \dots$ are the elements of A , listed in increasing order. Since A is a subset of $\{m_1, m_2, \dots\}$, for every natural number k , there exist some natural number l such that $n_k = m_l$. Since n_k is the k -th largest integer in A and since A is a subset of $\{m_1, m_2, \dots\}$, we see that $l \geq k$. Thus, $|a_{n_k}| = |a_{m_l}| < 1/2^l < 1/2^k$ (since $l \geq k$). Since $a_{n_k} \geq 0$, we have $0 \leq a_{n_k} = |a_{n_k}| < 1/2^k$. Now, using the Comparison Test, we see that since the series $\sum_{k=1}^{\infty} 1/2^k$ converges, so does the series $\sum_{k=1}^{\infty} a_{n_k}$.

On the other hand, suppose B is infinite. Let $n_1 < n_2 < \dots$ be the elements of B , listed in increasing order. Then, once again, by the same argument as in the previous paragraph, we see that $|a_{n_k}| \leq 1/2^k$ for all k . Since $a_{n_k} < 0$ for all k , we have the inequality $0 < -a_{n_k} < 1/2^k$. Using the Comparison Test, we see that the series $\sum_{k=1}^{\infty} (-a_{n_k})$ converges. Thus, the series $\sum_{k=1}^{\infty} a_{n_k}$ also converges. $\square$

Remark: The above proof gets considerably shortened if we use the Absolute Convergence Test. Indeed, if $|a_{m_k}| < 1/2^k$, we see that the series $\sum_{k=1}^{\infty} |a_{m_k}|$ converges, and hence so does the series $\sum_{k=1}^{\infty} a_{m_k}$. Thus, the length of the proof is actually half of what is given above. The additional argument involving the sets A and B is only needed above because the Absolute Convergence Test had not been discussed in class before this examination.

Problem 4. (5 points) Let $(x_n)_{n \in \mathbb{N}}$ be a sequence and let L be a real number. Suppose that every subsequence of $(x_n)_{n \in \mathbb{N}}$ has a subsequence which converges to L . Prove that $\lim_{n \in \mathbb{N}} x_n = L$. (Warning: Read the second sentence carefully.)

Solution. We prove this by induction. Suppose that it is *not true* that $\lim_{n \rightarrow \infty} x_n = L$. (This could mean that the sequence does not converge, or that if it converges, it converges to some other point.)

Recall that we say that $\lim_{n \rightarrow \infty} x_n = L$ if for any $\epsilon > 0$, there exists an integer N such that if $n > N$, we have $|x_n - L| < \epsilon$. Thus, if the sequence does not converge to L , there exists some $\epsilon > 0$ such that there is no integer N for which $|x_n - L| < \epsilon$ for all $n > N$. We fix such an ϵ . Then, for any integer N , there exists $n > N$ such that $|x_n - L| > \epsilon$.

We choose a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ such that $|x_{n_k} - L| > \epsilon$ for all k , by induction on k as follows:

First, choose $n_1 > 0$ such that $|x_{n_1} - L| > \epsilon$ (i.e. take $N = 0$ in the above statement). Now, suppose $x_{n_1}, \dots, x_{n_k}$ have been chosen. Then, choose n_{k+1} to be such that $n_{k+1} > n_k$ such that $|x_{n_{k+1}} - L| > \epsilon$ (i.e. take $N = n_k$ in the above statement). This gives us a sequence with the required property.

By construction, we see that $(x_{n_k})_{k \in \mathbb{N}}$ cannot have any subsequence converging to L since none of the terms of this sequence lie in the interval $(L - \epsilon, L + \epsilon)$. This contradicts the hypothesis. Thus, we must have $\lim_{n \rightarrow \infty} x_n = L$. $\square$

Problem 5. (5 points) Let $(x_n)_{n \in \mathbb{N}}$ be a sequence such that $x_n \geq 0$ for all n . Let $(y_n)_{n \in \mathbb{N}}$ be the sequence defined by $y_n = (-1)^n x_n$ for all n . If the sequence $(y_n)_{n \in \mathbb{N}}$ converges, prove that $(x_n)_{n \in \mathbb{N}}$ also converges.

Solution. Suppose $\lim_{n \rightarrow \infty} y_n = L$. Then, every subsequence of this sequence must converge to L (see Lecture 10, Theorem 6). Consider the two subsequences $(x_{2n})_{n \in \mathbb{N}}$ and $(x_{2n+1})_{n \in \mathbb{N}}$. Both these sequences converge to L .

Since $x_{2n} \geq 0$ for all n , we have $\lim_{n \rightarrow \infty} x_{2n} \geq 0$. Thus, $L \geq 0$. Similarly, since $x_{2n+1} \leq 0$ for all n , we get $\lim_{n \rightarrow \infty} x_{2n+1} \leq 0$. Thus, $L \leq 0$. Thus, $L = 0$.

Let $\epsilon > 0$ be given. Since $\lim_{n \rightarrow \infty} y_n = 0$, there exists an integer N such that if $n > N$, we have $|y_n - 0| < \epsilon$. However, $y_n = (-1)^n x_n$. Thus, $|y_n| = |x_n|$. Thus, $|x_n - 0| = |y_n - 0|$ for all n . Thus, we see that for $n > N$, we also have $|x_n - 0| = |x_n| = |y_n| < \epsilon$. Thus, we conclude that the sequence $(x_n)_{n \in \mathbb{N}}$ converges to 0 as well. $\square$